

ScanNet-SG: A Large-Scale Dataset for 3D Scene Graph Alignment

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Abstract—3D scene graph alignment establishes correspondences between graphs built from partially overlapping observations, enabling robots to maintain consistent object-level representations across time and agents. It arises in two complementary settings: frame-to-scan (F2S), for aligning partial observations to a global map, and subscan-to-subscan (S2S), for aligning independently built submaps. Existing datasets support only small-scale S2S alignment with limited object diversity and without vision–language representations, leaving large-scale, open-set benchmarks for both F2S and S2S largely unexplored. In this work, we propose an automated annotation pipeline that constructs open-set 3D scene graphs from RGB-D images and poses by integrating foundation models with point cloud processing tools. Applying this pipeline to ScanNet, we build ScanNet-SG, a large-scale benchmark for 3D scene graph alignment. ScanNet-SG contains over 700k alignment samples and covers 509 object categories from ScanNet labels and over 3k categories from GPT-4o-based tagging. Each object node is enriched with semantic labels, BERT embeddings, vision–language features, object point clouds, and 3D bounding boxes. By providing large-scale, multimodal data and supporting both F2S and S2S settings, ScanNet-SG provides a comprehensive benchmark for training and evaluating robust 3D scene graph alignment in open-world environments. Our dataset is released at: <https://github.com/tud-amr/ScanNet-SG>

I. INTRODUCTION

3D scene graphs [1] are emerging as a powerful representation for embodied agents operating in open-world environments, as they provide a compact and scalable abstraction that integrates semantic, geometric, and spatial information about objects. A typical 3D scene graph models the environment as nodes corresponding to objects and edges encoding relationships between them. This structured representation has demonstrated increasing value in a wide range of robotics tasks, including open-world navigation [2]–[5], mobile manipulation [6]–[8], simultaneous localization and mapping (SLAM) [9]–[11], and higher-level scene understanding and reasoning [12], [13].

In many of these applications, robots must continuously integrate observations across time and across agents by revisiting previously observed places and maintaining a consistent object-level representation of the environment. This requires identifying shared objects despite viewpoint changes and partial overlap. To fully leverage scene graphs in such deployments, robots must address the scene graph alignment problem, which involves establishing correspondences between

nodes in two scene graphs constructed from partially overlapping observations of the same physical environment [14]. In practice, alignment arises in two complementary settings: frame-to-scan (F2S) alignment, which aligns a partial observation (e.g., a single RGB-D frame) to a global scan-level scene graph under limited visible context, and subscan-to-subscan (S2S) alignment, which aligns two partial reconstructions (e.g., subscans / submaps) captured from different viewpoints, time periods, or agents. F2S is particularly useful for scene understanding and object relocation when a robot revisits a place or resumes operation after charging or recovery, and must quickly align a partial observation to its global map. S2S is particularly useful for global map fusion, allowing a single robot or multiple agents to align and merge independently built submaps.

Some recent works [11], [14]–[16] have started to formalize this problem, demonstrating scene graph alignment as a key primitive for single or multiple robot mapping and navigation. However, current datasets do not yet provide large-scale, open-set object alignment data with sufficient object diversity. Several datasets with 3D scene graph annotations have been proposed in recent years, such as 3D SSG [17], SG-FRONT [12], VLA-3D [18], and SG3D [19]. These datasets construct scene graphs on top of semantically annotated point cloud scans for navigation and mobile manipulation tasks. However, they do not contain scene graph pairs constructed from partially overlapping observations of the same environment, and thus do not support the scene graph alignment task.

Few datasets explicitly supporting 3D scene graph alignment have been proposed. Building upon the scene graph annotations of 3DSSG, SGAligner [14] introduces a benchmark for aligning partially overlapping 3D scene graphs. In this dataset, subscans are generated from globally aligned scans in 3RScan [20] with controlled overlap ratios. Each subscan forms a sub-scene graph, and nodes within overlapping regions are treated as ground-truth correspondences. The benchmark contains approximately 15.1k training pairs and 1.9k validation pairs, covering around 160 object classes. SG-Reg [11] further proposes a smaller dataset (approximately 4k samples) tailored for S2S alignment.

Although both SGAligner [14] and SG-Reg [11] provide object-level point clouds and semantic labels, they do not include vision–language representations, which we identify as a critical cue for robust real-world robotic deployment and a meaningful prior for scene-graph matching. Moreover, these datasets focus exclusively on S2S alignment. The F2S setting—where a partial observation must be matched to a larger global map, as required for object-level relocation—

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tion—remains unexplored. In contrast, our dataset is substantially larger, covering over 500 and 3k object classes across different groups and containing more than 700k samples in total. It additionally incorporates GroundingDino [21] vision–language features for each node and supports both S2S alignment and F2S matching, enabling evaluation in more realistic open-world scenarios.

In this work, we first propose a data annotation pipeline that can generate open-set 3D scene graphs from RGB-D images and poses by combining several foundation models with point cloud processing tools. We apply this pipeline to ScanNet, which contains more than 1,500 scans of 807 scenes, and build a dataset that supports both F2S and S2S alignment tasks. Each object node is associated with rich attributes, including semantic labels, BERT embeddings, VLM embeddings, object point clouds, and 3D bounding boxes. We then benchmark several state-of-the-art (SOTA) methods, including both learning-based and zero-shot approaches, on the two tasks.

II. SCENE GRAPH BUILDING AND DATASET CONSTRUCTION

This section first introduces the pipeline used to construct 3D scene graphs. We apply this pipeline to the ScanNet [22] dataset to build ScanNet-SG, a dataset designed for scene graph alignment in both the F2S and S2S tasks. We then describe the dataset construction and grouping strategies for the F2S and S2S tasks, respectively.

A. Scene Graph Building

Our 3D semantic scene graph is constructed from RGB-D images with given camera poses. The overall process consists of five steps:

1) *Image tagging*: Given an RGB image, we first use either the human annotations provided in ScanNet or an object-tagging model, such as ChatGPT-4o [23], to generate labels or textual descriptions of the objects present in the image.

2) *Segmentation and VLM feature extraction*: The generated text descriptions are provided as prompts to Grounded-SAM [24] to obtain instance-level object masks. Meanwhile, we acquire visual–language embeddings f_{vl} by extracting a 256-dimensional embedding vector from the hidden layers of Grounding DINO [21], which is used within Grounded-SAM. Our experiments demonstrate that this embedding is crucial for accurate alignment.

3) *Text embedding*: Text embeddings f_t are extracted from the object descriptions using SBERT [25], resulting in 384-dimensional embedding vectors.

4) *Bounding box computation*: Using the depth image and instance masks, we compute instance-aware point clouds. Oriented 3D bounding boxes are then estimated using PCA-based OBB fitting [26], [27]. The geometric feature f_g corresponds to the normalized size of the bounding box, where normalization is performed with respect to the maximum extent of the entire point cloud. The center of each bounding box is used as the spatial position of the corresponding object.

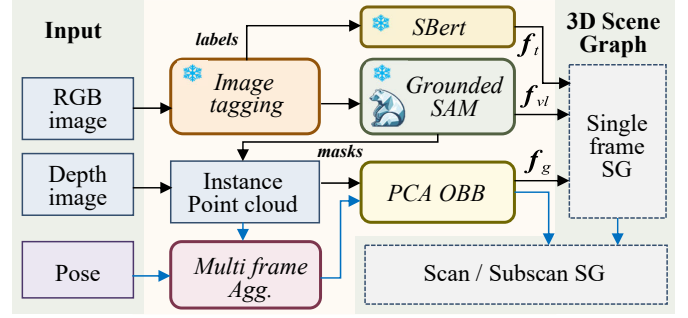


Fig. 1. Illustration of our object feature extraction and scene graph construction pipeline. Blue arrows indicate additional steps applied when building scene graphs from multiple frames, such as scans or subscans.

For a 3D scene graph constructed from a single frame, Steps 1)–4) yield the object nodes. Edges are established by computing pairwise distances between nodes and connecting each node to its N nearest neighbors within a distance threshold d_{th} . For scene graphs constructed from scans or subscans comprising multiple frames, the following additional step is applied:

5) *Multiview fusion*: We use instance masks and depth images from multiple views to generate per-instance point clouds. Instances corresponding to the same object are merged based on 3D mIoU and f_t similarity. During merging, the visual–language embeddings f_{vl} and text embeddings f_t are averaged, while the point clouds from different views are aggregated and downsampled using a voxel grid filter. The bounding box is then recomputed from the merged point cloud to obtain f_g . We further present the details of the fusion steps in Appendix A.

The key procedures and data flow is shown in figure 1. Our dataset is built upon ScanNet [22], which contains over 1,500 RGB-D scans spanning 807 indoor scenes. Most scenes contain more than one scan captured from different camera trajectories. In the following, we assume that the scene graphs $\mathcal{G}^A$ and $\mathcal{G}^B$ are constructed using the above procedure. In the frame-to-scan node matching task, $\mathcal{G}^A$ represents a smaller graph built from a single frame, while $\mathcal{G}^B$ corresponds to a complete scan of the scene. In the subscan-to-subscan node matching task, both $\mathcal{G}^A$ and $\mathcal{G}^B$ are constructed from incomplete scans of the same scene and share partial overlap. We present the details in the following.

B. Frame-to-Scan (F2S) Matching

To support frame-to-scan matching, we generate two groups of frame–scan pairs. *Self Scan* Group refers to the setting where the frame-level scene graph and the scan-level scene graph originate from the same scan. This setting evaluates whether a robot revisiting a previously observed pose, with access to only a single-frame partial observation, can correctly associate objects with their more complete counterparts in the fused map. *Cross Scan* Group considers frames sampled from a different scan of the same scene. This setting introduces novel viewpoints as well as newly appearing and missing regions, reflecting more realistic scenarios.

In the original ScanNet dataset, different scans of the same scene are represented in separate global coordinate frames, and object instances are not associated across scans. To enable cross-scan matching, we register the scan-level point clouds from different scans of ScanNet using RANSAC-based global registration followed by ICP refinement [28], [29] to estimate the transformation between scans. To ensure estimation quality, we manually inspect all registration results and discard scan pairs with noticeable misalignment. We present the registration performance in Tab. II. Thresholds of 0.05 m and 0.1 m are used to compute the mean and median fitness, RMSE, and overlap ratio (OR), respectively. The instances across scans are then associated using 3D mIoU and semantic labels, as described in Step 5).

We use Scenes 0–599 (1,276 scans) to construct the training set and Scenes 600–705 (237 scans) for testing. Frames are sampled every three frames from the original sequences. In total, the dataset contains approximately 557.0k training pairs and 87.9k test pairs, where 48.3% are generated from *Self Scan* and 52.7% from *Cross Scan*. The objects in this dataset span 509 unique semantic labels, and therefore we refer to this benchmark as *ScanNet-SG-509*.

To further increase object diversity and improve open-world generalization, we additionally use GPT-4o-mini [23] for image-based object tagging in Step 1). Compared to the original ScanNet annotations, the labels generated by GPT-4o-mini capture a substantially richer set of fine-grained and previously unannotated objects, including frequently occurring everyday items such as cable, mouse, water bottle, and light switch, as well as many other categories. Although only Scenes 0–99 are used for training and Scenes 601–610 and 696–705 for testing—amounting to 17.0% of the total data in *ScanNet-SG-509*—this process yields 3,195 unique object labels, which is 6.3× more than in *ScanNet-SG-509*. We refer to this benchmark as *ScanNet-SG-GPT*.

Fig. 2 and 3 present some examples of the scene graphs built with human annotated object labels provided in ScanNet and with ChatGPT-4o tagged object labels, respectively.

C. Subscan-to-subscan (S2S) Matching

Each subscan is constructed from 50 to 300 randomly selected yet temporally contiguous frames processed in Section II-B (150 to 900 frames in the original ScanNet dataset). We use the method described in Section II-A to build the point-cloud map and the scene graph for each subscan. If two subsamples from different scans of the same scene overlap, we include the pair as one dataset sample. In total, the dataset contains 13.4k training samples from Scenes 0–599 and 3.4k test samples from Scenes 600–705.

The samples in the S2S task have the following **difference** compared to the F2S task: (i) In F2S, the node-overlap ratio is typically close to one ($> 90\%$ on average) from $\mathcal{G}_A$ to $\mathcal{G}_B$ while very low ($< 20\%$ on average) from $\mathcal{G}_B$ to $\mathcal{G}_A$, since the frame-level nodes are usually a subset of those in the complete scan, except when new regions are observed in the cross-scan setting. In S2S, the overlap ratio between two subsamples spans (0, 1]. We report the detailed distribution on our website. (ii)

TABLE I
SUBSET STATISTICS OF SCANNet-SG DATASET

Task	Subset Name	Label Source	Training Samples	Test Samples
F2S	<i>ScanNet-SG-509</i>	ScanNet	557.0k	118.6k
	<i>ScanNet-SG-GPT</i>	gpt-4o	87.9k	26.9k
S2S	<i>ScanNet-SG-Subscan</i>	ScanNet	13.4k	3.4k

TABLE II
REGISTRATION PERFORMANCE FOR CROSS-SCAN MATCHING

THR	Mean Fitness $\uparrow$	Med. Fitness $\uparrow$	Mean RMSE $\downarrow$ (cm)	Med. RMSE $\downarrow$ (cm)	Mean OR $\uparrow$	Med. OR $\uparrow$
0.05 m	0.750	0.798	2.29	2.30	0.800	0.840
0.10 m	0.844	0.899	3.21	3.10	0.898	0.938

In F2S, the two scene graphs are expressed in the camera frame and the world frame, respectively, and therefore differ substantially in their coordinate systems. In contrast, in S2S, both scene graphs are expressed in world frames. While these world frames are not identical because the graphs are built from different scans, they share a similar gravity alignment (i.e., the z -axis points upward). (iii) In ScanNet-SG-509, F2S matches a small graph with 4.5 nodes on average to a larger graph with 22.2 nodes on average, whereas S2S matches two graphs with 17.3 nodes on average.

The general data statistics are presented in Table I. More statistics and visualizations of the dataset can be found on our website.

III. RESULTS

This section first presents the experimental setup and then reports the results of baseline methods on the F2S and S2S evaluation tasks, respectively.

A. Experimental Setup

Five baseline methods are compared for F2S and S2S tasks. As a learning-based baseline, we include SG-Reg [11], a state-of-the-art scene-graph alignment method that leverages instance point clouds, 3D bounding boxes, and text features through a dedicated matching network. We further consider four zero-shot baselines. The first method is ROMAN [10], which performs geometrically consistent matching using point cloud features and VLM embeddings. For a fair comparison, we adapt ROMAN by replacing its original FastSAM [30] segments with our object-level segments and by using VLM embeddings generated from Grounded-DINO [21] rather than CLIP [31]. The remaining three zero-shot baselines rely on cosine similarity between object embeddings: VLM embeddings from Grounded-DINO [21], language embeddings from SBERT [25], and the concatenation of both. For each query object, the instance with the highest similarity exceeding a predefined threshold is selected as the match. These VLM and language embeddings are trained to yield high cosine similarity for semantically consistent objects while separating distinct ones, making cosine similarity a strong yet simple

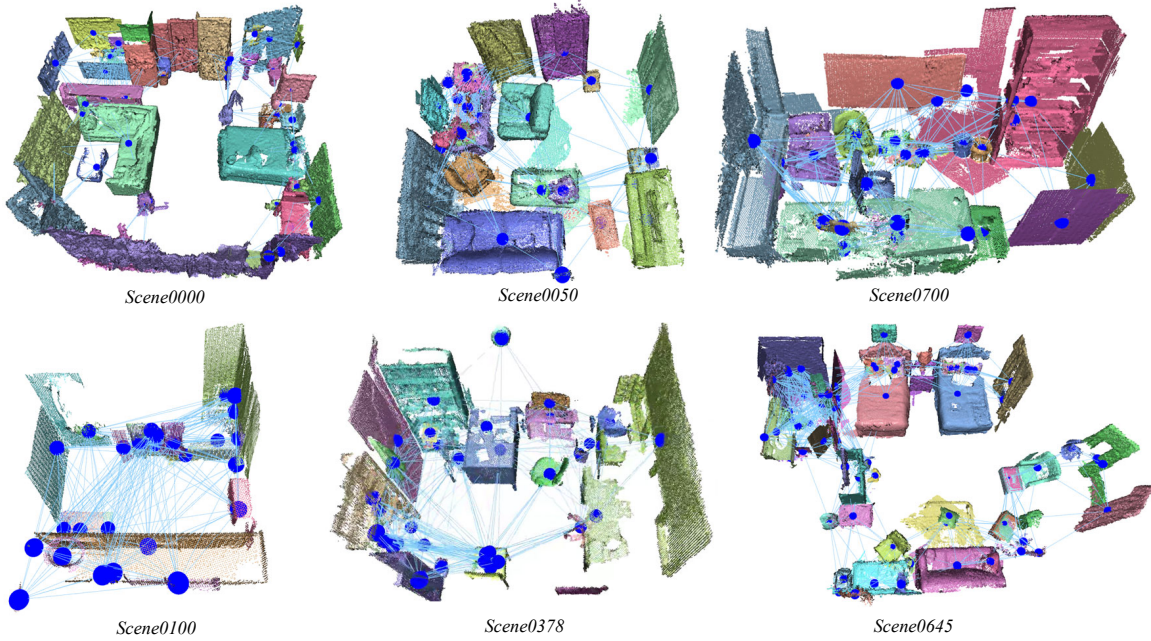


Fig. 2. Examples of 3D scene graphs constructed using ScanNet human-annotated object labels (*ScanNet-SG-509*). The blue points show the centers of objects. An edge is added to connect two objects if their distance is smaller than two meters.

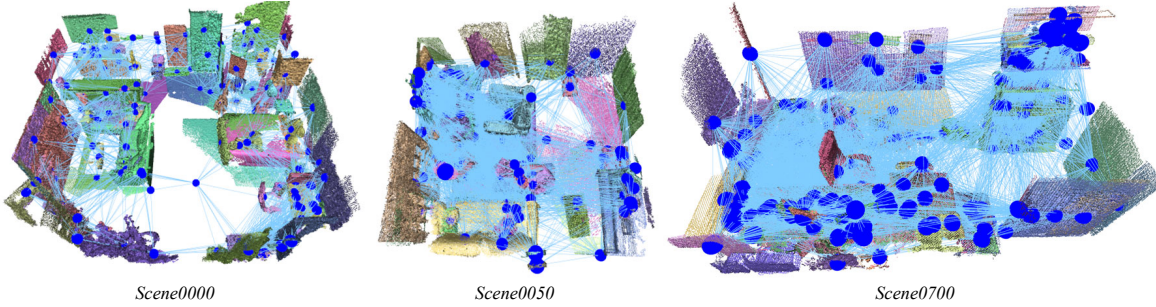


Fig. 3. Examples of 3D scene graphs constructed using GPT-4o-tagged objects *ScanNet-SG-GPT*. The three scenes correspond to those shown in the first row of Fig. 2, but contain substantially more object instances and a significantly larger set of object classes.

baseline for zero-shot matching. The thresholds and other key parameters of the compared methods are optimized with Optuna [32] to achieve the highest F1 score.

All models are trained on NVIDIA A40 GPUs and tested on a desktop equipped with an NVIDIA RTX 3080 Ti. Each model is trained for 10 epochs using the AdamW optimizer [33] with a learning rate of 2×10^{-4} and a batch size of 64. The results are evaluated with four metrics: **Accuracy**, **Precision**, **Recall**, and **F1** score. Each metric is computed per test sample over the nodes and then averaged over the test set.

B. Frame-to-scan Node Matching

For this task, all models are trained on the *ScanNet-SG-509* training split, which includes a mixture of self-scan and cross-scan samples. We evaluate performance on three test sets: (i) the self-scan group, (ii) the cross-scan group, and (iii) the *ScanNet-SG-GPT* subset. These test sets represent progressively more challenging settings. In (i), query frames are drawn from the same scan used to construct the map,

resulting in minimal viewpoint and coverage variation. In (ii), query frames originate from different scanning trajectories, leading to larger viewpoint changes. In (iii), we further evaluate on *ScanNet-SG-GPT*, whose expanded object vocabulary generated by GPT-4o presents the most challenging generalization setting. Importantly, we do not use the *ScanNet-SG-GPT* training split; instead, all models are trained exclusively on *ScanNet-SG-509*. This design enables a direct assessment of cross-vocabulary generalization from *ScanNet-SG-509* to *ScanNet-SG-GPT*,

Table III–V summarize the F2S matching results across the three test sets. Figure 4 illustrate the results of two methods on seven different matching samples. A key observation is the strong impact of pretrained semantic features for open-vocabulary instance matching. The cosine-similarity baselines demonstrate that VLM embeddings and SBERT embeddings already provide a surprisingly strong signal for correspondence. In particular, VLM CosSim and VLM+Bert CosSim consistently outperform the learning-based baseline

TABLE III
MATCHING RESULTS IN SCANNet-SG-509 SELF SCAN GROUP (SIMPLE)

Method	Acc $\uparrow$	Pr $\uparrow$	Re $\uparrow$	F1 $\uparrow$
SG-Reg [11]	0.114	0.158	0.190	0.172
ROMAN [10]	0.433	0.590	0.484	0.519
VLM CosSim [21]	0.731	0.754	0.939	0.811
Bert CosSim [25]	0.512	0.515	0.877	0.616
VLM+Bert CosSim [21], [25]	0.762	0.766	0.973	0.836

TABLE IV
MATCHING RESULTS IN SCANNet-SG-509 CROSS SCAN GROUP (MEDIUM)

Method	Acc $\uparrow$	Pr $\uparrow$	Re $\uparrow$	F1 $\uparrow$
SG-Reg [11]	0.027	0.039	0.042	0.040
ROMAN [10]	0.358	0.414	0.374	0.380
VLM CosSim [21]	0.540	0.550	0.800	0.620
Bert CosSim [25]	0.455	0.458	0.715	0.519
VLM+Bert CosSim [21], [25]	0.564	0.562	0.876	0.656

SG-Reg [11] and ROMAN [10] across all three test sets. Combining VLM and SBERT embeddings further improves performance compared to using either modality alone, indicating that the two representations capture complementary semantic information.

Across the three evaluation settings, performance degrades as task difficulty increases. In the self-scan group (Simple), VLM+Bert CosSim achieves the highest F1 score (0.836), followed closely by VLM CosSim (0.811). In the cross-scan group (Medium), all methods experience performance drops due to viewpoint changes and partial overlap, yet VLM+Bert CosSim remains the strongest baseline with 0.656 F1. The most challenging setting, ScanNet-SG-GPT (Difficult), introduces a substantially larger and more diverse object vocabulary. Here, all methods show significant degradation, but VLM+Bert CosSim again achieves the highest F1 (0.462), demonstrating better generalization to expanded open-vocabulary scenarios.

SG-Reg performs poorly across all F2S settings. Its reliance on instance point clouds and coordinate-dependent geometric encoding appears insufficient under frame-to-scan conditions, where coordinate-system discrepancies and partial observations are pronounced. ROMAN achieves moderate performance but consistently trails behind the cosine-similarity baselines, suggesting that its geometric consistency constraints and PCA-based point cloud features are less effective than direct semantic similarity in this task.

Overall, the F2S results highlight that pretrained semantic embeddings—particularly VLM features—play a dominant role in open-vocabulary frame-to-scan node matching, while geometry-centric approaches struggle under large viewpoint and coordinate variations.

C. Subscan-to-subscan Node Matching

Table VI reports the results of the S2S node matching task. The results of two high-performing baseline methods and three of our variants are further illustrated in Fig. 5.

TABLE V
MATCHING RESULTS IN SCANNet-SG-GPT TEST SET (DIFFICULT)

Method	Acc $\uparrow$	Pr $\uparrow$	Re $\uparrow$	F1 $\uparrow$
SG-Reg [11]	0.012	0.019	0.033	0.024
ROMAN [10]	0.143	0.187	0.156	0.160
VLM CosSim [21]	0.306	0.354	0.504	0.381
Bert CosSim [25]	0.260	0.339	0.377	0.306
VLM+Bert CosSim [21], [25]	0.356	0.363	0.740	0.462

TABLE VI
SUBSCAN-TO-SUBSCAN MATCHING RESULT

Method	Acc $\uparrow$	Pr $\uparrow$	Re $\uparrow$	F1 $\uparrow$
SG-Reg [11]	0.056	0.171	0.078	0.107
ROMAN [10]	0.430	0.505	0.268	0.336
VLM CosSim [21]	0.461	0.370	0.906	0.502
Bert CosSim [25]	0.535	0.382	0.751	0.477
VLM+Bert CosSim [21], [25]	0.518	0.400	0.898	0.530

In the S2S setting, ROMAN exhibits relatively high precision (0.505) but substantially lower recall (0.268), resulting in the lowest F1 among the cosine-based methods. This suggests that its additional geometric constraints may limit match proposals, leading to conservative but incomplete matching. SG-Reg [11] remains significantly behind all other methods. We observe that in the training set, SG-Reg achieves a precision of 0.501 and a recall of 0.512, indicating a tendency to overfit to training data.

Among all baselines, VLM+Bert CosSim achieves the highest overall F1 score (0.530), reflecting a better balance between precision and recall compared to single-modality embeddings. VLM CosSim achieves the highest recall but at the cost of lower precision. Bert CosSim shows improved accuracy compared to VLM CosSim but slightly lower F1 due to reduced recall.

Overall, the S2S results confirm that pretrained VLM and BERT embeddings provide strong performance in subscan-level matching. However, purely similarity-based approaches struggle to maintain precision in complex scenes. In comparison, ROMAN, which considers geometric consistency between two graphs, achieves higher precision.

IV. CONCLUSION

This paper introduced a benchmark for 3D scene graph alignment under partial observations, covering both frame-to-scan (F2S) and subscan-to-subscan (S2S) node matching. The benchmark evaluates performance under varying levels of viewpoint change, overlap, and open-vocabulary expansion, enabling systematic comparison of baseline methods. The results indicate that pretrained semantic embeddings, particularly VLM-based representations, provide strong signals for open-vocabulary correspondence. Nevertheless, the overall performance of existing methods remains limited, especially in cross-scan and open-vocabulary settings, where substantial drops in accuracy and F1 score reveal a clear gap toward robust and reliable alignment. The current dataset is limited by simplified edge semantics, where relations are mainly represented by geometric distance and a single “next to”

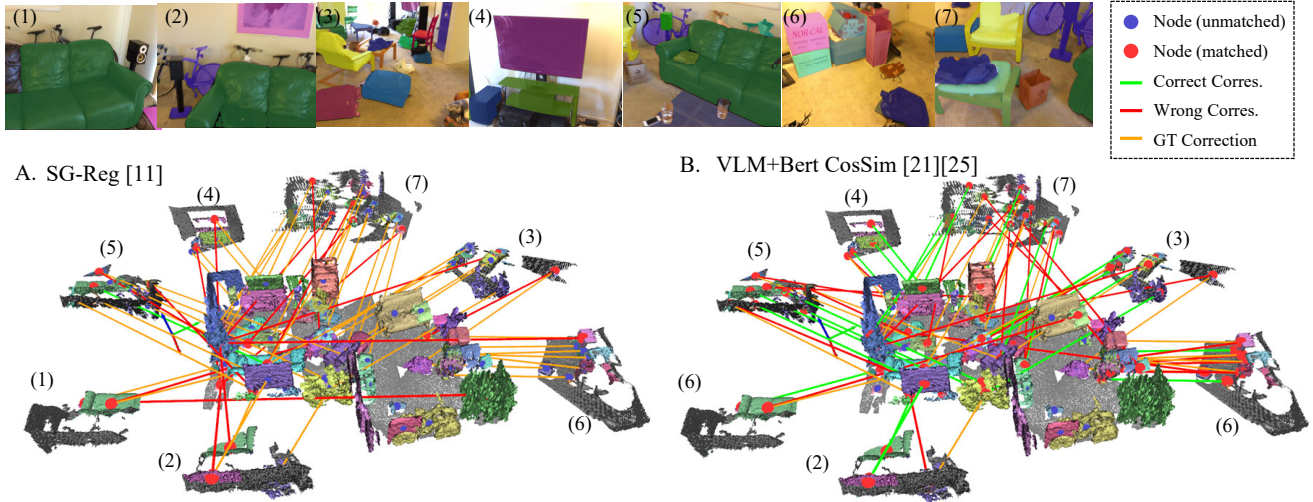


Fig. 4. Visualization of matching results produced two methods. In each subplot, the central point cloud corresponds to the full scan of *Scene0673_00*. The query frames are selected from a separate scan of the same scene, *Scene0673_01* (cross-scan), and are shown in the top-left corner as instance-segmented RGB images labeled (1)–(7), corresponding to ScanNet frame IDs 15, 105, 252, 639, 1050, 1521, and 1830 [22]. Objects are visualized with distinct colors, and each object center is marked with a blue dot (unmatched) or a red dot (matched). Correct correspondences are indicated by green lines, while incorrect correspondences are shown in red. Orange lines denote the ground-truth corrected correspondences for either incorrect or missed matches. For visualization purposes, each frame point cloud (originally in its camera coordinate system) is transformed into the world coordinate system of *Scene0673_00* and then translated to a nearby location around the scan point cloud. This transformation is applied only for visualization and is not provided to any method.

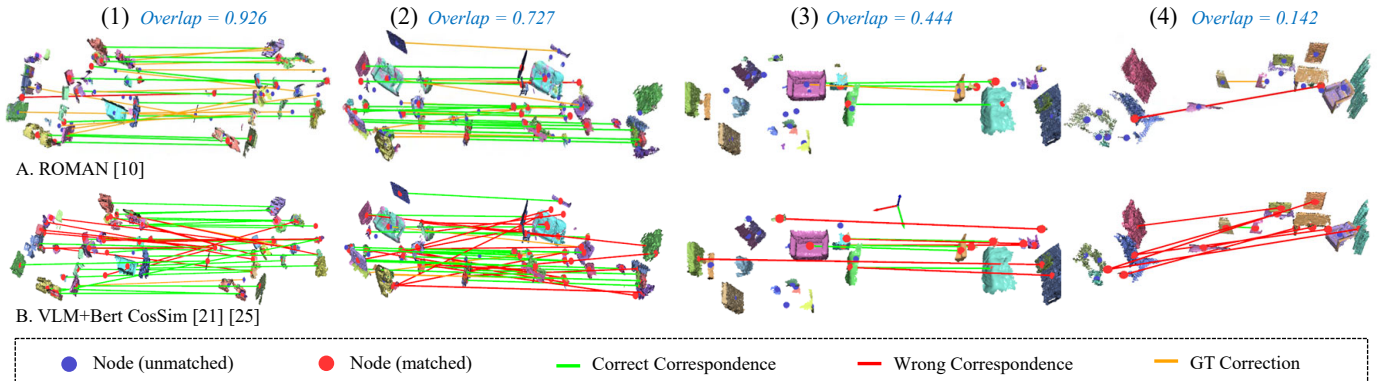


Fig. 5. Results on the S2S alignment task. The subscan pairs in (1)–(4) are: (1) *Scene0673_00* frames 1383–2139 and *Scene0673_01* frames 1584–1989; (2) *Scene0673_00* frames 654–1140 and *Scene0673_02* frames 234–1032; (3) *Scene0698_00* frames 732–942 and *Scene0698_01* frames 837–1302; and (4) *Scene0642_00* frames 168–777 and *Scene0642_01* frames 1890–2367. The node overlap ratio between two subscans is shown with blue text on the left. The overlap ratio decreases from 0.926 to 0.142 from (1) to (4). Points belonging to different instances are rendered in random colors. Background point clouds are omitted for clarity.

description. Future work will extend relation diversity and expand the benchmark to larger-scale environments.

APPENDIX A DETAILS OF MULTIVIEW FUSION

A key technique in multiview fusion is instance tracking across input frames. When ScanNet provides semantic labels and coarse instance segmentations projected from the 3D mesh with manually assigned instance IDs, as in the construction of *ScanNet-SG-509*, we associate fine-grained instances obtained from GroundedSAM with ScanNet coarse instances using IoU-based matching to inherit consistent instance IDs. Points belonging to the same instance are then merged and downsampled using a voxel grid filter with a resolution of 0.02 m.

For open-set labels tagged by GPT-4o, where no ground-truth instance IDs are available, we perform instance tracking based on both semantic and geometric consistency. Specifically, two instances are considered identical if their SBERT embedding cosine similarity and their voxel-level 3D IoU both exceed predefined thresholds. SBERT embeddings help distinguish semantically different objects that may have high geometric overlap, such as a tablecloth on a table or a book on a laptop. This matching condition is first applied sequentially across frames to construct an initial instance-aware point cloud map for each scan. We then iteratively compare all existing instances and merge those satisfying the same condition until convergence. To improve efficiency, a center distance check is applied prior to similarity evaluation to filter unlikely candidates. This post-merging step is particularly important

for resolving duplicate instances caused by multiple times of scans of the same object.

Although GroundedSAM leverages the foundation model SAM for segmentation, segmentation noise near object boundaries can introduce outlier points in the fused map. To address this issue, we further apply DBSCAN [34] and Statistical Outlier Removal (SOR) filtering, retaining only the largest connected point cluster for each instance to improve the quality of the reconstructed object point clouds.

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